

Repeated Exam MA-224, part 2

Course code:	MA-224
Course name:	Discrete mathematics 2
Date:	23. February 2024
Time:	09:00 – 15:00 (6 hours)
Number of pages:	5
Number of tasks:	12
Permitted equipment:	All written or printed material (including local digital documents and resources on your own computer). Calculator. No communication is allowed during the exam.
Remarks:	Present your answers so that it is clear what methods you have used, and how you have used them. Your answer should contain a calculation or an argument explaining how the answer was reached; otherwise, it is incomplete. All problems 1-12 are scored equally. If a problem has more than one part, then each part is scored equally within the problem.

Problem 1

We consider the following theorem.

$$\text{Th1: } [(p \rightarrow r) \wedge (q \rightarrow r)] \iff [(p \vee q) \rightarrow r]$$

(a) Prove Th1 using truth tables.

(b) Prove Th1 using logical inferences.

For your convenience, here are some possible inference rules.

equivalence between $\rightarrow$ and $\wedge$: $A \rightarrow B \iff \neg A \vee B$

deMorgan's Law for $\wedge$: $\neg(A \wedge B) \iff \neg A \vee \neg B$

deMorgan's Law for $\vee$: $\neg(A \vee B) \iff \neg A \wedge \neg B$

commutativity of $\wedge$: $A \wedge B \iff B \wedge A$

associativity of $\wedge$: $(A \wedge B) \wedge C \iff A \wedge (B \wedge C)$

distributivity of $\wedge$: $(A \vee B) \wedge C \iff (A \wedge C) \vee (B \wedge C)$

commutativity of $\vee$: $A \vee B \iff B \vee A$

associativity of $\vee$: $(A \vee B) \vee C \iff A \vee (B \vee C)$

distributivity of $\vee$: $(A \wedge B) \vee C \iff (A \vee C) \wedge (B \vee C)$

Problem 2

Define the following sequence recursively: $s_i = 4 \cdot (i + 2)$.

Problem 3

Consider the following EBNF grammar with the start symbol s . Terminals are enclosed in single quotes ('go'). Alternatives are denoted by '|'.

$$s \rightarrow \text{'go'} \mid \text{'stop'} \mid \text{'while'} s \text{'do'} s \mid s \text{' ; ' } s$$

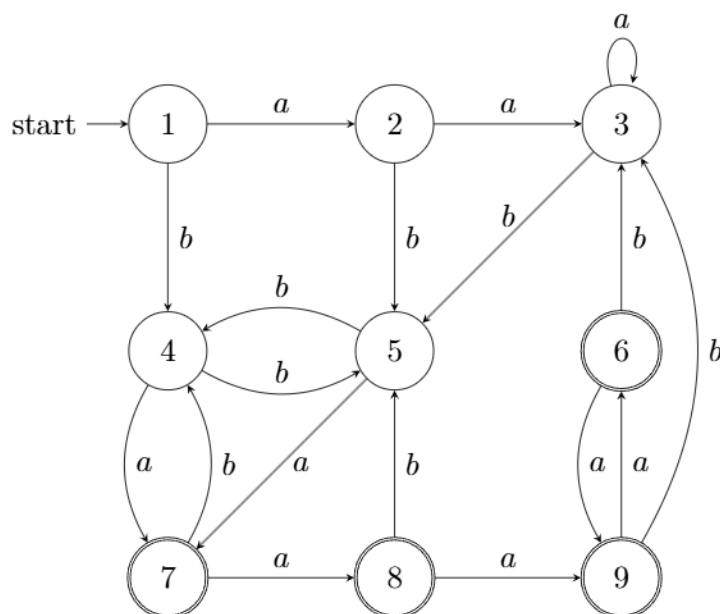
(a) Find a derivation in the EBNF grammar above for the string

while stop do go ; stop.

(b) Is the grammar ambiguous? Give reasons.

Problem 4

Minimize the following DFA.



Problem 5

Is the code below correct related to the following specification?

Specification: $\text{SQRT}(x, \text{res}:\text{NUM}) \text{ res} := \text{res}' \mid \text{res}' * \text{res}' = x$

Code: `float sqrt(x:float) { return (-1) * Math.sqrt(x); }`

Problem 6

We define a relation $=_{nin3}$ between Norwegians such that two Norwegians are related with $=_{nin3}$ if they have the same first three digits of their national identity number. Otherwise, they are not related with $=_{nin3}$.

Prove that $=_{nin3}$ is an equivalence relation, i.e. that $=_{nin3}$ is reflexive, symmetric, and transitive.

You can assume that there is a function $nin3$, which extracts the first three digits of the national identity number (i.e., a number) for each Norwegian N .

Problem 7

Consider words of length $k > 0$ using only the letters 0 and 1. We require that we cannot write two 1's next to each other. For example, when $k = 1, 2$ and 3, the following table lists all the allowed words:

$k = 1$	$k = 2$	$k = 3$
0	00	000
1	10	100
	01	010
		001
		101

Let a_k be the number of allowed words of length k . From the table we see that $a_1 = 2$, $a_2 = 3$ and $a_3 = 5$.

Show that a_k satisfies the recurrence relation

$$a_k = a_{k-1} + a_{k-2} \quad \text{for all } k > 2.$$

Problem 8

Let H_1 and H_2 both be subgroups of an Abelian group $(G, \cdot)$. Prove that, in general, the union

$$H = H_1 \cup H_2$$

is not a subgroup of G by giving a counterexample.

Problem 9

Let $\phi(n) = |P_n|$ be the Euler totient function.

- Prove by giving a counterexample that $\phi(n)$ is not an increasing function of n .
- Prove using the definition of $\phi(n)$ that $\phi(2^k) = 2^{k-1}$, for every positive integer $k \geq 1$.

Problem 10

Let G be a group of order 33 and let $H \subset G$ be a subgroup such that $\{1\} \neq H \neq G$. Use Lagrange's theorem to prove that the index $[G : H]$ is a prime number.

Problem 11

Let the encoding function

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be given by multiplication with a generator matrix:

$$E(w_1 \ w_2 \ w_3) = (w_1 \ w_2 \ w_3) \cdot \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

- Prove that E is able to detect all transmission errors of weight 1.

- (b) Give an example showing that E is unable to correct all transmission errors of weight 1.

Problem 12

Given the (3×6) generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Assume that a 3-bit message is encoded using G and transmitted over a noisy channel, and that the received message is $r = \mathbf{011001}$. Assume that at most one bit error occurs during the transmission of the message.

Compute the syndrome of r and use it to detect and correct any error in the transmission.